

73-103 · CMU-Q

# Principles of Macroeconomics

Fall 2026

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## Meeting 4

# Measuring Prices

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Wed 2 Sep · CORE 4.2–4.4

How we measure the cost of living — CPI, the deflator, and what a price index can and cannot do.

# This meeting we are in the Prices cell — only that cell

*Description · Tools · Use. CPI / deflator, real vs nominal.*

|                  | Topic                     | Description                             | Tools                                  | Use                              |
|------------------|---------------------------|-----------------------------------------|----------------------------------------|----------------------------------|
| Measurement      | <b>GDP</b>                | What the economy produced               | Circular flow, $Y \equiv C+I+G+NX$     | Read a release; Qatar energy GDP |
|                  | <b>Prices</b>             | Persistent rise in the price level      | CPI / deflator, real vs nominal        | Contract, wage, peg              |
|                  | <b>Labor</b>              | Employment, unemployment, participation | Labor-force identities; short AI/labor | GCC hiring / participation       |
| Short-run demand | <b>Goods / multiplier</b> | Demand determines Y                     | ZZ, multiplier, Blanchard 3            | Fiscal impulse                   |
|                  | <b>Financial markets</b>  | How i is set                            | Money market, Blanchard 4              | QCB / Fed; $i = \bar{i}$         |
|                  | <b>IS-LM</b>              | Goods + financial together              | IS, LM, policy mix, dashboards         | Shock → which curve, which way   |
| Medium run       | <b>Phillips</b>           | Inflation and unemployment together     | Phillips curve, Blanchard 7–8          | Wage / inflation tradeoff        |
|                  | <b>Growth</b>             | Why levels differ                       | Production function, growth accounting | Handout 4 + R                    |
| Open             | <b>Open economy</b>       | Trade, rates, and the peg               | NX, Blanchard 17–19                    | Shock → NX / peg (Qatar)         |

today

*Midterm · Debate*

## MOTIVATION

# Why measuring prices matters

### Driving question

**How would you put a country's inflation in context — high, or extremely high?**

# Motivation

Argentina, then Qatar

WORLD | AMERICAS

## Argentina's Inflation Surges After New President Cuts Subsidies

Javier Milei is attempting to fix the poor economy he inherited with pro-market shock therapy

By [Ryan Dubé](#) [Follow](#) and [Silvina Frydlewsky](#)

Updated Jan. 11, 2024 at 2:35 pm ET

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Argentine economy

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## Javier Milei's next economic mission: affordable air fryers

Libertarian leader scraps tariffs to break open Argentina's protectionist economy, unnerving domestic industry

Ciara Nugent in Buenos Aires JANUARY 7 2025

[Comments](#) 237 [Print](#)

Javier Milei has begun wrenching open Argentina's protectionist economy, slashing tariffs in a bid to beat down high prices on everything from air fryers to electronics.

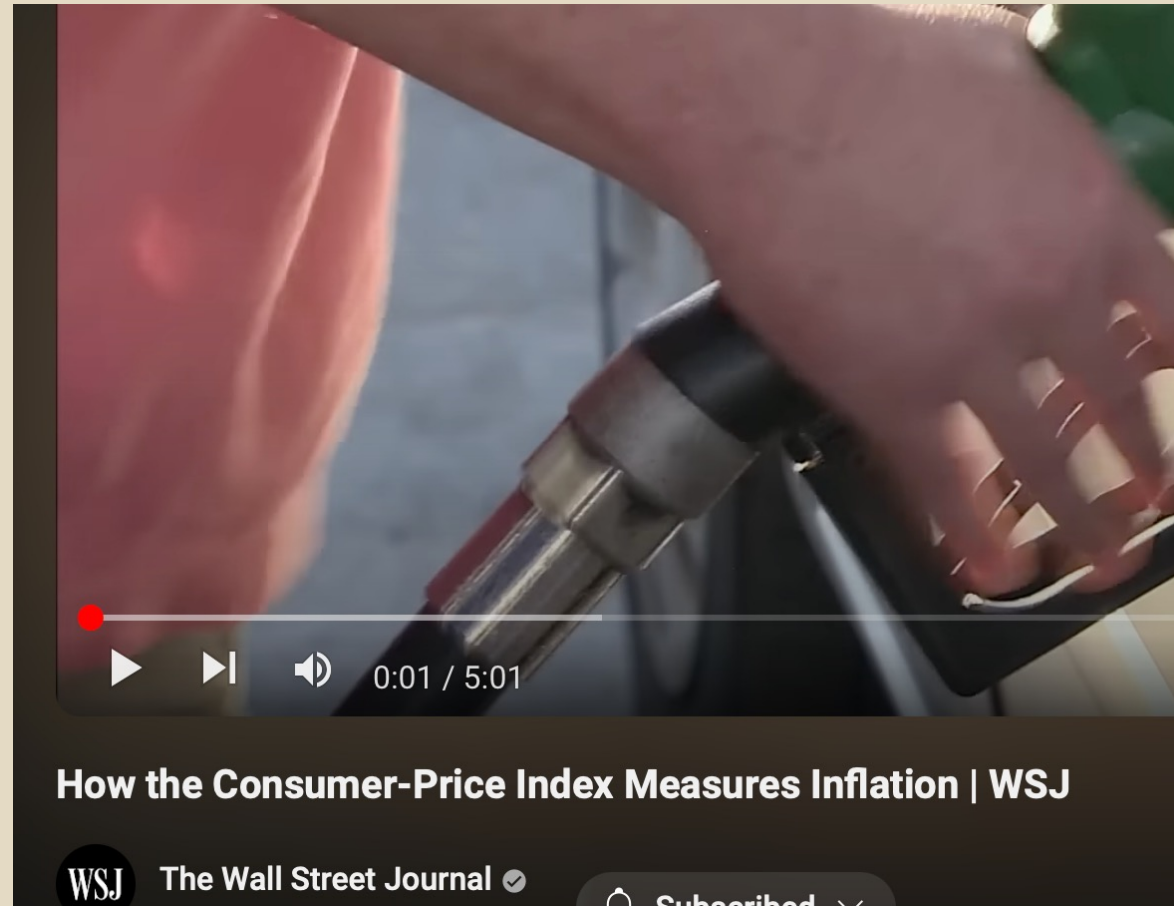
Argentines are ordering from Amazon for the first time and supermarkets are beginning to stock their shelves with Tide laundry detergent and Ecuadorean tinned tuna, as the libertarian president dismantles a web of duties and regulations that have made many imported products near unaffordable.

A Black & Decker air fryer that sells for \$100 in the US retails for \$289 in [Argentina](#), according to listings on online retailers. A Zara slip dress that costs

How would you put Argentina's inflation in context? Look up the same statistics for Qatar. What trade-offs might anti-inflation policy create for consumers, firms, and the government?

# How is inflation measured?

*WSJ: How the Consumer-Price Index Measures Inflation*



<https://www.youtube.com/watch?v=oRdLvp6H3CU>

## OUTLINE

# What we cover this meeting

|    |                                                      |
|----|------------------------------------------------------|
| 01 | Causes of inflation                                  |
| 02 | Measuring the cost of living                         |
| 03 | What's the Consumer Price Index (CPI)                |
| 04 | How is it calculated?                                |
| 05 | What is it used for?                                 |
| 06 | What are the issues measuring the CPI?               |
| 07 | Alternatives to the CPI                              |
| 08 | Using the CPI to compare dollar amounts across years |
| 09 | Important considerations                             |

## CAUSES OF INFLATION

# Why the price level rises

### Driving question

**Which channel is at work — costs, demand, expectations, or money?**

# Causes of inflation

## Cost-push

Higher input costs (energy, wages, imported intermediates) push the price level up.

## Demand-pull

Spending outruns what the economy can produce at current prices.

## Expectations

If people expect inflation, they set wages and prices accordingly — and the expectation can stick.

## Monetary (QTM)

More money chasing the same goods. Quantity theory of money.

$$M \times V = P \times Q$$

M = money supply   V = velocity   P = price level   Q = real output. If V and Q are relatively stable, a rise in M shows up in P. Fisher, Hume, Friedman. Keynesian caveat: unused capacity / liquidity trap. Structuralist caveat: supply bottlenecks.



HOW IS INFLATION MEASURED?

## Three related price indexes

**Driving question**

**What does each index actually price — a consumer basket, that basket without food and energy, or what firms buy?**

# The Consumer Price Index (CPI)

*Three different yet related price indexes*

## 1 • Consumer Price Index

Overall cost of goods and services bought by an average consumer. Tracks the cost of living over time.

## 2 • Core CPI

Same as CPI, but excluding food and energy — the volatile items.

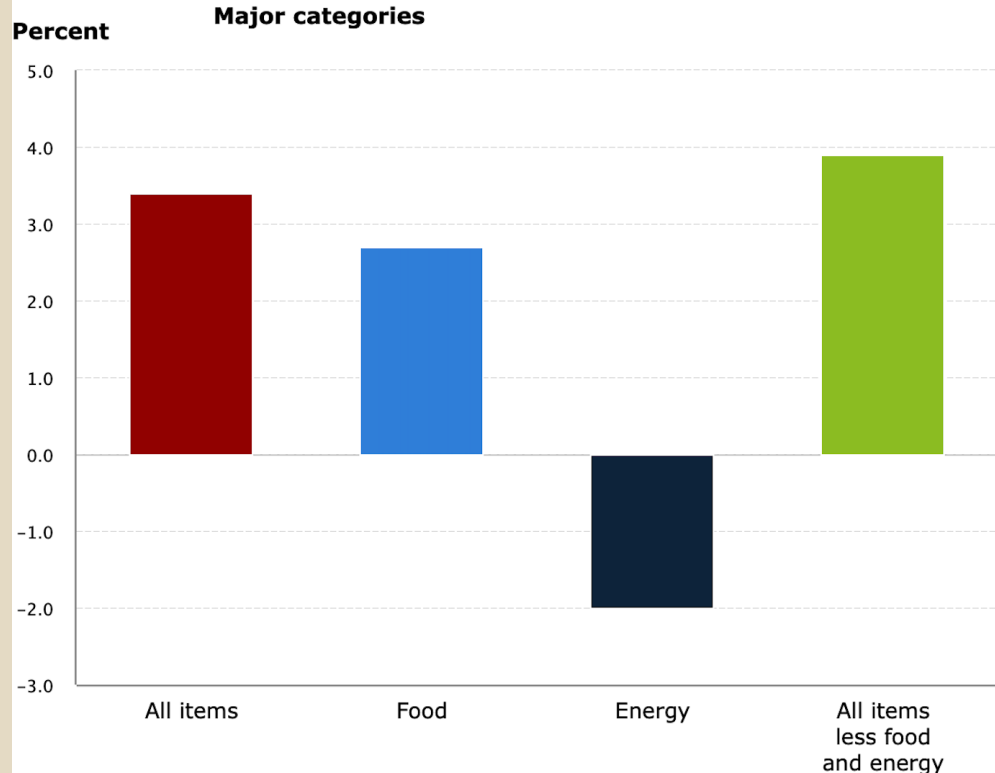
## 3 • Producer Price Index

Cost of goods and services bought by firms. A leading indicator for the CPI.

# Important considerations

12-month percentage change, Consumer Price Index, selected categories, December 2023, not seasonally adjusted

[Click on columns to drill down](#)



Why do you think this information is useful?

PPI for final demand, 12-month percent change, not seasonally adjusted



Source: U.S. Bureau of Labor Statistics.

PPI is a leading indicator for CPI. Why?

## CONSTRUCTION

# How is this CPI constructed?

### Driving question

**Fix a basket, collect prices, compute the cost, choose a base year — then the inflation rate follows.**

## Example 1 — two-good basket

*10 baklavas and 5 umm ali. Base year 2019.*

| Year | Price of baklava | Price of umm ali | Cost of basket                           |
|------|------------------|------------------|------------------------------------------|
| 2019 | \$12             | \$18             | $\$12 \times 10 + \$18 \times 5 = \$210$ |
| 2020 | \$14             | \$20             | $\$14 \times 10 + \$20 \times 5 = \$240$ |
| 2021 | \$16             | \$22             | $\$16 \times 10 + \$22 \times 5 = \$270$ |

### CPI in each year (2019 base)

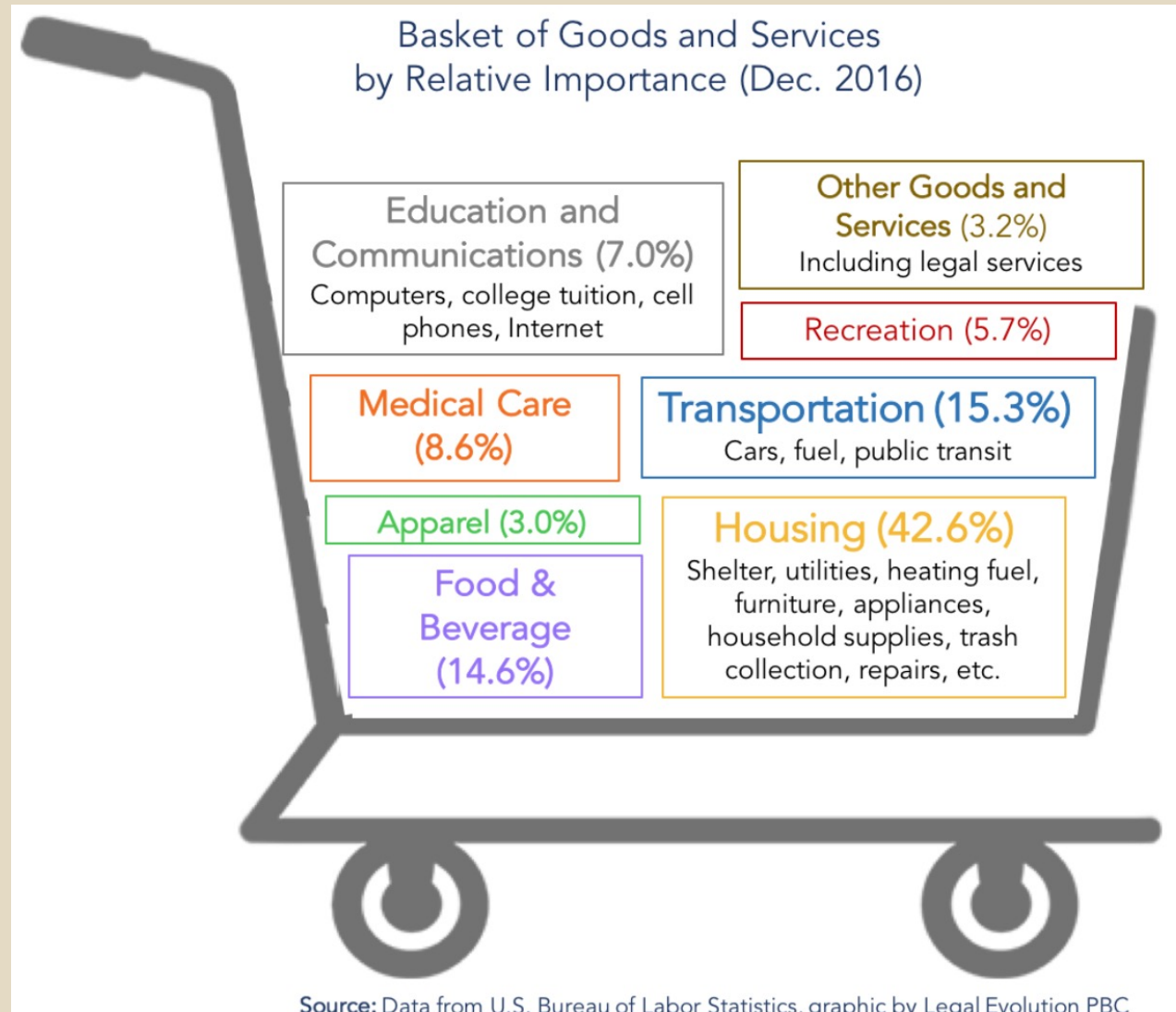
$$2019: 100 \times (\$210 / \$210) = 100$$

$$2020: 100 \times (\$240 / \$210) = 114.3$$

$$2021: 100 \times (\$270 / \$210) = 128.6$$

# What's a typical basket of goods and services?

U.S. relative importance, December 2016



# What are the most significant contributors in Qatar?

*National Planning Council, quarterly CPI, third level*

**Table 3: Quarterly Consumer Price Index, Third level**

| Goods and Services Groups                                        | Weight       | 2022-Q2       | 2022-Q3       | 2022-Q4       | 2023-Q1       | 2023-Q2       |
|------------------------------------------------------------------|--------------|---------------|---------------|---------------|---------------|---------------|
| <b>GENERAL PRICE INDEX (CPI)</b>                                 | <b>10000</b> | <b>102.68</b> | <b>104.16</b> | <b>106.84</b> | <b>105.46</b> | <b>106.48</b> |
| <b>FOOD AND NON-ALCOHOLIC BEVERAGES</b>                          | <b>1258</b>  | <b>105.57</b> | <b>108.19</b> | <b>107.66</b> | <b>106.58</b> | <b>105.59</b> |
| Food                                                             | 1133         | 106.11        | 108.97        | 108.07        | 106.72        | 105.49        |
| Beverages                                                        | 125          | 100.92        | 101.49        | 104.15        | 105.40        | 106.43        |
| <b>TOBACCO</b>                                                   | <b>27</b>    | <b>246.00</b> | <b>246.00</b> | <b>246.00</b> | <b>246.00</b> | <b>246.00</b> |
| Tobacco                                                          | 27           | 246.00        | 246.00        | 246.00        | 246.00        | 246.00        |
| <b>CLOTHING AND FOOTWEAR</b>                                     | <b>511</b>   | <b>91.30</b>  | <b>92.70</b>  | <b>94.28</b>  | <b>94.49</b>  | <b>93.25</b>  |
| Clothing                                                         | 435          | 89.11         | 91.05         | 92.62         | 92.89         | 92.73         |
| Footwear                                                         | 76           | 104.15        | 102.44        | 104.05        | 103.93        | 96.29         |
| <b>HOUSING, WATER, ELECTRICITY, GAS, AND OTHER FUEL</b>          | <b>2189</b>  | <b>91.89</b>  | <b>94.78</b>  | <b>98.10</b>  | <b>98.09</b>  | <b>98.11</b>  |
| Actual Rental for Housing                                        | 1934         | 90.40         | 93.47         | 97.22         | 97.33         | 97.28         |
| Maintenance and repair of the Dwelling                           | 103          | 90.69         | 94.48         | 92.48         | 88.68         | 90.39         |
| Water Supply and Miscellaneous services Relating to the Dwelling | 61           | 108.00        | 108.00        | 108.00        | 108.00        | 108.00        |
| Electricity, Gas and Other Fuels                                 | 91           | 118.42        | 117.95        | 118.34        | 119.12        | 119.27        |
| <b>FURNISHINGS, HOUSEHOLD EQUIPMENT AND ROUTINE HOUSEHOLD</b>    | <b>770</b>   | <b>105.49</b> | <b>105.48</b> | <b>105.59</b> | <b>104.02</b> | <b>107.78</b> |
| Furnishing, Carpet and Other floor covering                      | 180          | 101.12        | 99.85         | 98.51         | 88.46         | 101.00        |
| Household Textiles                                               | 50           | 94.40         | 96.62         | 98.77         | 102.72        | 109.11        |
| Household Appliance                                              | 82           | 99.81         | 99.60         | 103.45        | 104.14        | 104.06        |
| Glassware, Tableware and Household Utensils                      | 41           | 90.00         | 103.00        | 102.00        | 102.00        | 99.00         |
| Tools and Equipment for House and Garden                         | 7            | 103.00        | 97.00         | 100.33        | 103.67        | 98.00         |
| Good and Services for Routine Household Maintenance              | 410          | 110.83        | 110.03        | 109.78        | 110.07        | 111.85        |
| <b>HEALTH</b>                                                    | <b>179</b>   | <b>98.27</b>  | <b>98.06</b>  | <b>98.52</b>  | <b>99.91</b>  | <b>99.66</b>  |
| Medical products, Appliance and Equipment                        | 31           | 100.13        | 102.44        | 101.34        | 99.03         | 98.38         |
| Outpatient Services                                              | 79           | 96.25         | 95.43         | 96.42         | 99.21         | 98.91         |
| Hospital Services                                                | 69           | 101.81        | 101.81        | 101.81        | 101.81        | 101.81        |

|                                                                 |             |               |               |               |               |               |
|-----------------------------------------------------------------|-------------|---------------|---------------|---------------|---------------|---------------|
| <b>TRANSPORT</b>                                                | <b>1459</b> | <b>109.14</b> | <b>108.80</b> | <b>109.75</b> | <b>111.92</b> | <b>110.40</b> |
| Purchase of Vehicles                                            | 659         | 114.33        | 113.67        | 114.00        | 117.67        | 113.33        |
| Operation of Personal Transport Equipment                       | 402         | 102.65        | 101.67        | 102.54        | 102.61        | 102.24        |
| Transport services                                              | 398         | 112.45        | 113.15        | 114.70        | 118.11        | 117.62        |
| <b>COMMUNICATION</b>                                            | <b>587</b>  | <b>93.06</b>  | <b>91.08</b>  | <b>91.49</b>  | <b>89.49</b>  | <b>104.17</b> |
| postal Services                                                 | 1           | 105.00        | 100.00        | 100.00        | 100.00        | 100.00        |
| Telephone and Telefax Equipment                                 | 145         | 88.00         | 85.00         | 96.00         | 88.00         | 86.00         |
| Telephone and Telefax Services                                  | 441         | 93.53         | 91.67         | 90.98         | 89.60         | 106.07        |
| <b>RECREATION AND CULTURE</b>                                   | <b>1268</b> | <b>106.35</b> | <b>112.15</b> | <b>123.82</b> | <b>112.12</b> | <b>116.88</b> |
| Audio-Visual, Photographic and Information Processing Equipment | 76          | 88.26         | 94.26         | 89.10         | 93.64         | 87.50         |
| Other recreational Items and Equipment Gardens and Pets         | 84          | 105.11        | 106.84        | 104.59        | 106.22        | 110.19        |
| Recreation and Culture services                                 | 54          | 98.31         | 98.79         | 95.36         | 95.72         | 93.72         |
| News Papers, Books and Stationary                               | 43          | 104.52        | 104.26        | 104.00        | 102.87        | 103.13        |
| Packages Holidays                                               | 1011        | 108.00        | 114.67        | 129.67        | 115.00        | 121.00        |
| <b>EDUCATION</b>                                                | <b>575</b>  | <b>110.80</b> | <b>111.23</b> | <b>112.03</b> | <b>115.45</b> | <b>115.30</b> |
| Pre-Primary and Primary education                               | 231         | 112.35        | 112.35        | 112.35        | 112.35        | 112.35        |
| Preparatory and Secondary education                             | 180         | 111.46        | 112.65        | 115.02        | 115.35        | 116.02        |
| NON-Tertiary Education                                          | 12          | 105.25        | 108.75        | 111.75        | 111.75        | 111.75        |
| Post-Secondary- Tertiary Education                              | 152         | 108.00        | 108.00        | 108.00        | 120.33        | 119.00        |
| <b>RESTAURANTS AND HOTELS</b>                                   | <b>608</b>  | <b>106.58</b> | <b>108.21</b> | <b>114.86</b> | <b>109.33</b> | <b>106.76</b> |
| Catering Services                                               | 577         | 106.25        | 108.40        | 108.64        | 107.98        | 107.21        |
| Accommodation Services                                          | 31          | 111.00        | 105.67        | 198.00        | 127.33        | 100.67        |
| <b>MISCELLANEOUS GOODS AND SERVICES</b>                         | <b>569</b>  | <b>110.60</b> | <b>107.46</b> | <b>107.27</b> | <b>109.07</b> | <b>109.76</b> |
| Personal Care                                                   | 339         | 102.36        | 102.69        | 102.55        | 104.35        | 102.54        |
| Personal Effects                                                | 183         | 130.84        | 119.17        | 118.77        | 122.47        | 129.00        |
| Insurance                                                       | 26          | 101.00        | 101.00        | 101.00        | 101.00        | 101.00        |
| Other Services                                                  | 21          | 117.00        | 113.00        | 113.00        | 106.00        | 106.00        |

Which goods and services contribute more to the CPI in Qatar? How do Qatar's basket weights compare with those of the average consumer in the US?

## ACTIVE LEARNING

# Calculating CPI and inflation

### Driving question

**Can you take a two-good basket from prices to an inflation rate?**



## Practice question 1 — calculate CPI and the inflation rate

### **CPI basket**

10 lbs of lamb

20 lbs of beef

**Base year: 2017**

| Year | Price of lamb | Price of beef |
|------|---------------|---------------|
| 2017 | \$3           | \$3           |
| 2018 | \$4           | \$4           |
| 2019 | \$8           | \$5           |

Calculate the CPI for all years.

What was the inflation rate from 2018 to 2019?

## Practice question 1 — solution

| Year | Lamb | Beef | Cost of basket                          | CPI   |
|------|------|------|-----------------------------------------|-------|
| 2017 | \$3  | \$3  | $\$3 \times 10 + \$3 \times 20 = \$90$  | 100.0 |
| 2018 | \$4  | \$4  | $\$4 \times 10 + \$4 \times 20 = \$120$ | 133.3 |
| 2019 | \$8  | \$5  | $\$8 \times 10 + \$5 \times 20 = \$180$ | 200.0 |

### Inflation rate, 2018–2019

$$(200.0 - 133.3) / 133.3 = 50.0\%$$

Same method as the baklava basket: cost relative to the 2017 base, then the percent change.

## PROBLEMS WITH THE CPI

# Three measurement problems

### Driving question

**Why might a rising CPI overstate the rise in the cost of living?**

# Problems with the CPI

*There are three problems in measuring the CPI*

**01**

## **Substitution bias**

Households switch away from goods that have become more expensive.

**02**

## **Introduction of new goods**

New products raise the value of a dollar; a fixed basket misses that.

**03**

## **Unmeasured quality change**

Better goods mean each dollar buys more; quality is hard to measure.

## **Problem 1 — substitution bias**

Consumers can substitute toward less expensive products in a given period.

**Why is this a problem in measuring the CPI?**

A fixed basket cannot capture that substitution.

**Implication**

CPI overestimates the cost of living (most likely).

## **Problem 2 — introduction of new goods**

New goods increase variety and let consumers find products that more closely meet their needs. Dollars become more valuable.

The CPI misses this effect because it uses a fixed basket of goods.

### **Implication if new products rise**

CPI overestimates increases in the cost of living.

## **Problem 3 — unmeasured quality change**

Improvements in the quality of goods in the basket increase the value of each dollar.

Statistical agencies try to account for quality changes but probably miss some, as quality is hard to measure.

### **Implication if quality rises**

CPI overestimates increases in the cost of living.

# Same sticker price, not the same phone

A quality-change example. Release price \$799 in both years.

|                      | iPhone 13 (2021)              | iPhone 14 (2022)                                                         |
|----------------------|-------------------------------|--------------------------------------------------------------------------|
| <b>Release price</b> | \$799 (base model)            | \$799 (base model)                                                       |
| <b>Processor</b>     | A15 Bionic (4-core GPU)       | A15 Bionic (5-core GPU); better cooling                                  |
| <b>Camera</b>        | Dual 12MP (wide + ultra-wide) | Improved main sensor; Photonic Engine; better low-light and front camera |
| <b>Battery</b>       | ~19 hours video playback      | Slightly longer (~20 hours video)                                        |
| <b>Other</b>         | Standard capabilities         | Emergency SOS via satellite; Crash Detection                             |



## ALTERNATIVES TO THE CPI

# Other ways to track the price level

### Driving question

**If CPI has known biases, what do we use instead — and why does the Fed prefer PCE?**

# Alternatives to the CPI

## CPI vs. PCE: Untangling the Alphabet Soup of Inflation Gauges

Fed's preferred measure of consumer-price growth may not be the one you think, but both have their place



By Jo Craven McGinty [Follow](#)

March 20, 2015 at 12:03 pm ET

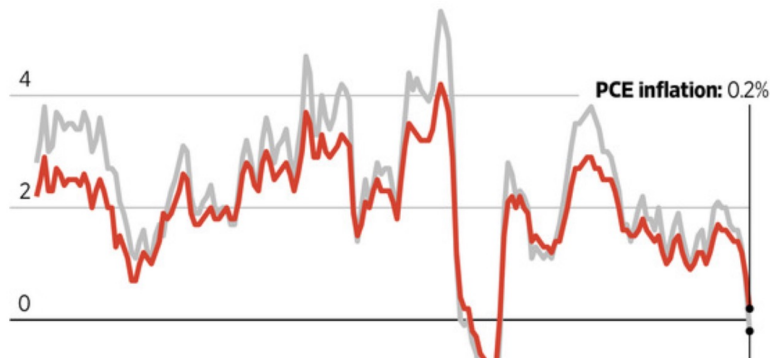
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### The Low Down

The U.S. Federal Reserve prefers the personal consumption expenditures price index to the more familiar consumer-price index to gauge inflation.

6% change from a year earlier



### Alternative 1 · PCE

Personal Consumption Expenditures Price Index. Broader, chained, monthly basket.

### Alternative 2 · Truflation

A high-frequency alternative index.

### Alternative 3 · GDP deflator

Prices of goods and services produced domestically.

What's the Federal Reserve, and why do they use the PCE instead of the CPI?

## ACTIVE LEARNING

# Case studies about inflation

### Driving question

**Can you pick the response that matches the price-index logic we just used?**

## Practice — case study 1

Hyperinflation. Which comprehensive response is most likely?

Businesses face escalating prices and a rapidly depreciating currency. Which strategic response is the most complete?

- A** Hold prices, raise inventory, lock long-term fixed-price supplier contracts, avoid credit.
- B** Raise prices daily, cut inventory, seek short-term variable-price contracts, rely heavily on credit.
- C** Cut prices, raise inventory to stimulate demand, long-term variable-price contracts, minimize credit.
- D** Adjust prices frequently, keep inventory minimal, short-term fixed-price supplier contracts, prefer cash to credit.

## Practice — case study 2

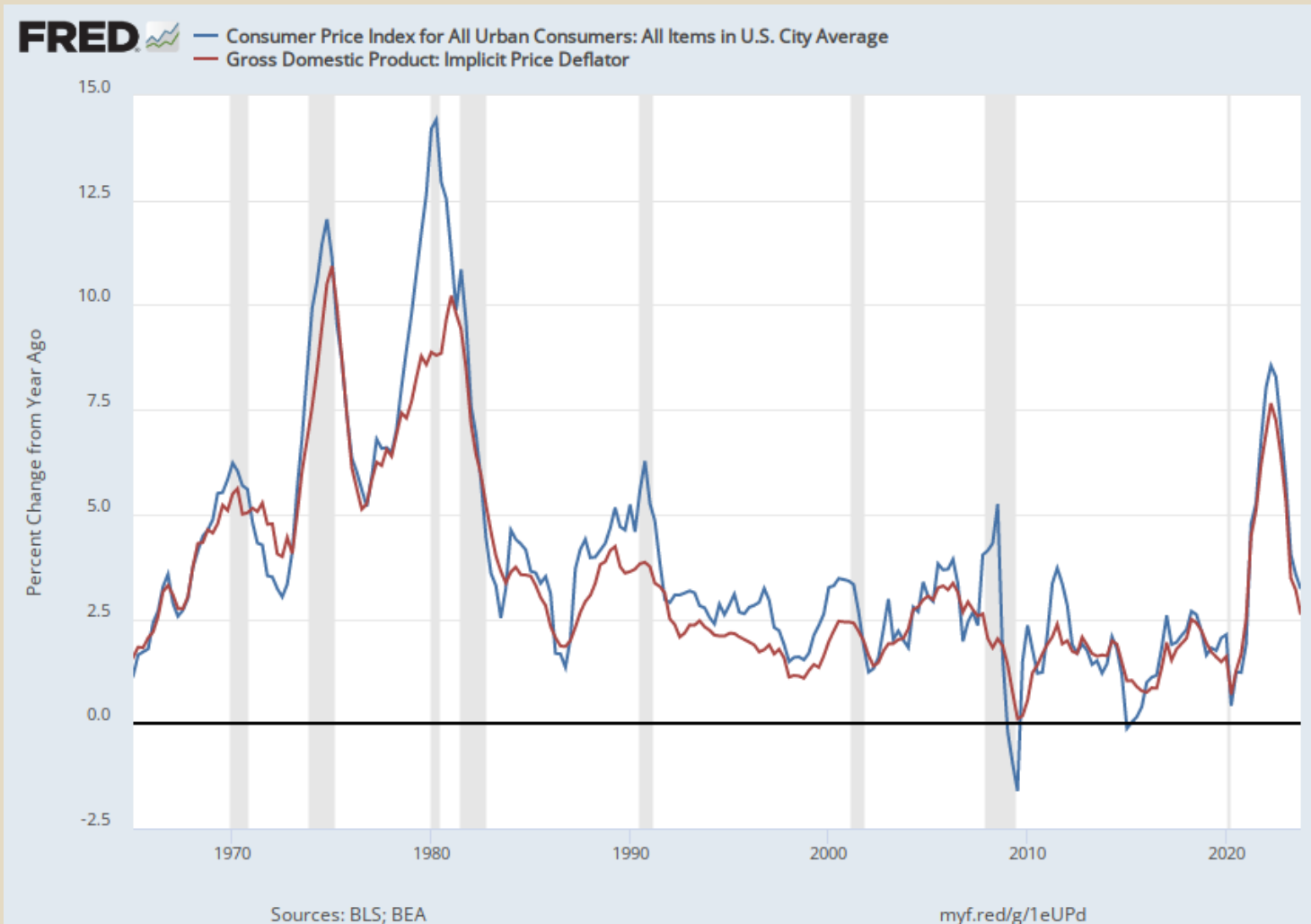
*PPI is rising in a developed economy. Which forecast is theoretically sound?*

Business analysts watch several indexes. Given a steady rise in the PPI, which forecast is most reasonable?

- A** The PPI rise will barely touch consumer prices and is unlikely to move CPI much.
- B** The PPI rise is a precursor to deflation and will pull CPI down.
- C** The PPI rise will likely pass through to consumer prices and push CPI up.
- D** The PPI rise reflects stronger activity and will cut consumer prices via more supply, so CPI falls.

# Alternative 4 — the GDP deflator

*Two indicators of inflation, 1965–2019*



The two series tend to move together, but...

Think about 2021.

Why does CPI change substantially more than the GDP deflator?

# CPI versus the GDP deflator

*Both track price changes. Two crucial differences.*

## They track different prices

GDP deflator → prices of all goods and services produced domestically.

CPI → prices of all goods and services bought by consumers.

Imported consumer goods: in CPI, not in the deflator. Capital goods produced at home: in the deflator, not in CPI.

## The basket

CPI: a fixed basket of goods and services bought by consumers.

GDP deflator: prices of goods and services currently produced domestically — the basket updates with production.

## Practice — CPI versus the GDP deflator

In each scenario, determine the effects on the CPI and the GDP deflator.

**A** Starbucks raises the price of muffins.

**B** Caterpillar raises the price of the industrial tractors it manufactures at its Illinois factory.

**C** Armani raises the price of the Italian jeans it sells in the U.S.



## Practice — CPI versus the GDP deflator

*Same three. Work them before the answers.*

**A** Starbucks raises the price of muffins.

**B** Caterpillar raises the price of the industrial tractors it manufactures at its Illinois factory.

**C** Armani raises the price of the Italian jeans it sells in the U.S.

# Answers

---

## A • Muffins

**The CPI and the GDP deflator both rise.**

Muffins are produced at home and bought by consumers.

## B • Caterpillar tractors

**The GDP deflator rises; the CPI does not.**

Tractors are produced in Illinois (capital goods), not a consumer good.

## C • Armani jeans

**The CPI rises; the GDP deflator does not.**

Italian jeans are in the U.S. consumer basket and are imports.

# Summary — CPI measurements and alternatives

## **Main indicator**

**Consumer Price Index (CPI)**

## **Variations**

Core CPI · Median CPI · 16% trimmed-mean CPI

## **Alternatives**

PCE · GDP deflator · Truflation US Inflation Index

# The disparate effects of inflation

ECONOMY | CONSUMERS | THE OUTLOOK

## Inflation Usually Hits Harder for Poor Families. For a Couple of Years, It Didn't.

New research on how inflation varies between the poor, middle class and rich paints a different picture of poverty and inequality

By [Justin Lahart](#) [Follow](#)

Aug. 25, 2024 at 5:30 am ET

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Should policymakers consider alternative inflation measures that better account for the experiences of different income groups when designing monetary or fiscal policies? Why or why not?

## CORRECTING VARIABLES

# Correcting economic variables for inflation

### Driving question

**How do you put a dollar amount from one year into the prices of another year?**

# Correcting economic variables for the effects of inflation

## Amount in today's dollars

$$\text{amount in year T dollars} \times (\text{CPI today} / \text{CPI in year T})$$

### Worked example — the tuition years

$$\text{Amount in 2018 dollars} = \text{amount in 1990 dollars} \times \text{CPI 2018} / \text{CPI 1990}$$

That is the conversion used on the next slides, with published tuition and CPI figures.

## Practice — comparing wages

**How would you compare a wage paid in one year with a wage paid in another year?**

Use the same conversion as the formula on the previous slide. Put both wages in the same year's dollars, then compare.

## Practice — comparing tuition increases

| Tuition and fees at U.S. colleges | 1990    | 2018     |
|-----------------------------------|---------|----------|
| Private non-profit 4-year         | \$9,340 | \$35,830 |
| Public 4-year                     | \$1,908 | \$10,230 |
| Public 2-year                     | \$906   | \$3,660  |
| CPI                               | 130.7   | 251.1    |

Express the 1990 tuition figures in 2018 dollars, then compute the percentage increase in real terms for all three types of schools. Which type experienced the largest increase in real tuition costs?



## Answers — fill the real columns

|                                    | 1990    | 2018     | % change |
|------------------------------------|---------|----------|----------|
| <b>CPI</b>                         | 130.7   | 251.1    | 92.1%    |
| <b>Private 4-year (current \$)</b> | \$9,340 | \$35,830 |          |
| <b>Private 4-year (in 2018 \$)</b> | ?       | ?        | ?        |
| <b>Public 4-year (current \$)</b>  | \$1,908 | \$10,230 |          |
| <b>Public 4-year (in 2018 \$)</b>  | ?       | ?        | ?        |
| <b>Public 2-year (current \$)</b>  | \$906   | \$3,660  |          |
| <b>Public 2-year (in 2018 \$)</b>  | ?       | ?        | ?        |

# Answers

|                                    | 1990     | 2018     | % change |
|------------------------------------|----------|----------|----------|
| <b>CPI</b>                         | 130.7    | 251.1    | 92.1%    |
| <b>Private 4-year (current \$)</b> | \$9,340  | \$35,830 |          |
| <b>Private 4-year (in 2018 \$)</b> | \$17,944 | \$35,830 | 99.7%    |
| <b>Public 4-year (current \$)</b>  | \$1,908  | \$10,230 |          |
| <b>Public 4-year (in 2018 \$)</b>  | \$3,665  | \$10,230 | 179.1%   |
| <b>Public 2-year (current \$)</b>  | \$906    | \$3,660  |          |
| <b>Public 2-year (in 2018 \$)</b>  | \$1,741  | \$3,660  | 110.3%   |

**Largest real increase: public 4-year colleges and universities.**

## Practice — case study 3

### Amazon and regional cost of living

A rigid, uniform compensation framework raised discontent where living costs are high — more quits, lower morale. Which approach should Amazon take?

- A** A universal raise, to reinforce equity across regions.
- B** A geographically nuanced model that factors in regional living costs.
- C** Richer non-monetary perks, without restructuring base pay.
- D** Standardize pay on role responsibilities only; ignore regional disparities.

## TRENDS

# Inflation by good and service, US 1998–2018

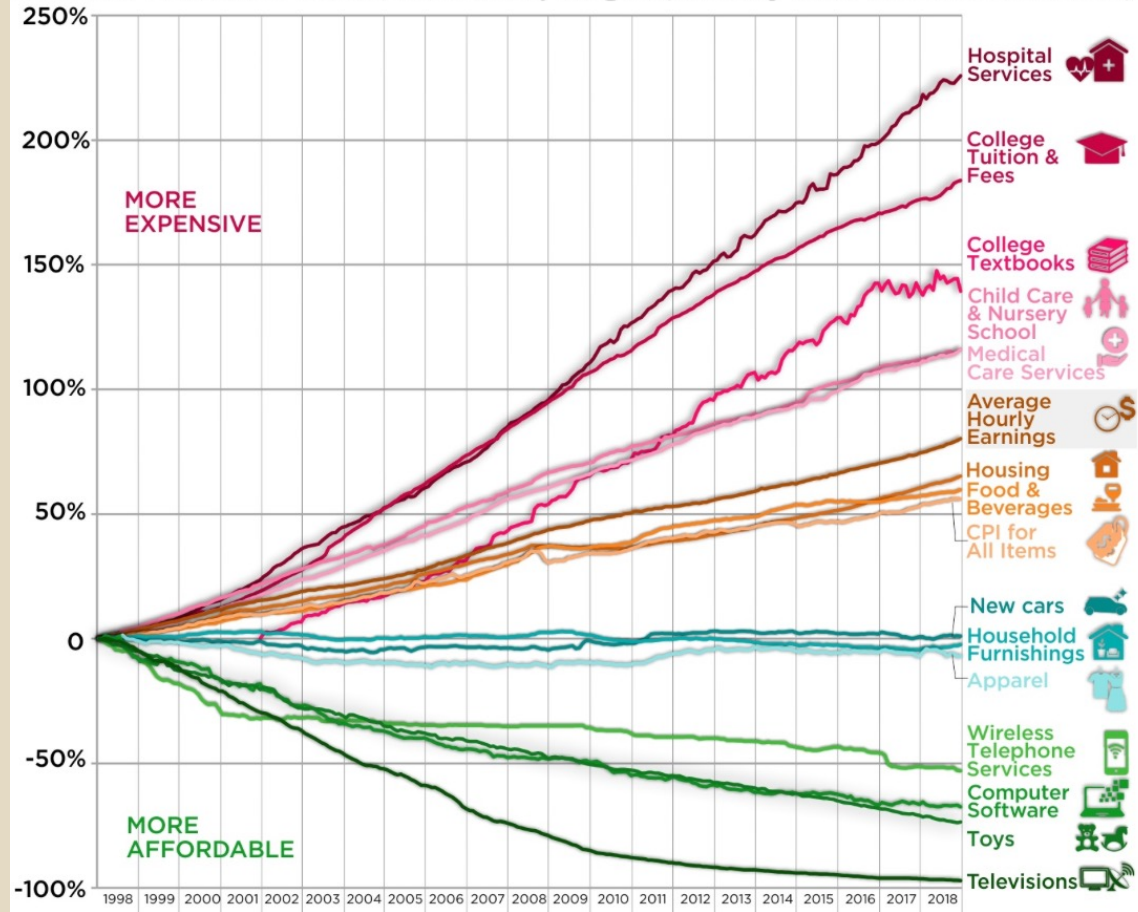
### Driving question

**Which prices ran ahead of wages — and which ran behind?**

# Inflation trends by goods and services

## 20 Years of Price Changes in The United States

Selected Consumer Goods & Services, Wages (January 1998 to December 2018)



Article & Sources:  
<https://howmuch.net/articles/price-changes-in-usa-in-past-20-years>  
CPI and other price indices - Bureau of Labor Statistics - <https://data.bls.gov/PDQWeb/cu>  
Average hourly earnings - Bureau of Labor Statistics - <https://data.bls.gov/timeseries/CES0500000008>

howmuch.net

What do you see in this graph?

What factors explain the divergent trends?

## IMPORTANT CONSIDERATIONS

# Three uses of a price index

### Driving question

**Indexation, regional prices, and the gap between nominal and real interest rates.**

# Important consideration I — indexation

**When some dollar amount is automatically corrected for inflation → indexation**

**01**

## **Long-term contracts**

Negotiated between unions and companies.

**02**

## **Social Security benefits**

Payments that move with a price index.

**03**

## **Income-tax brackets**

Thresholds corrected so inflation does not push people into a higher band.

# Important consideration II — regional differences

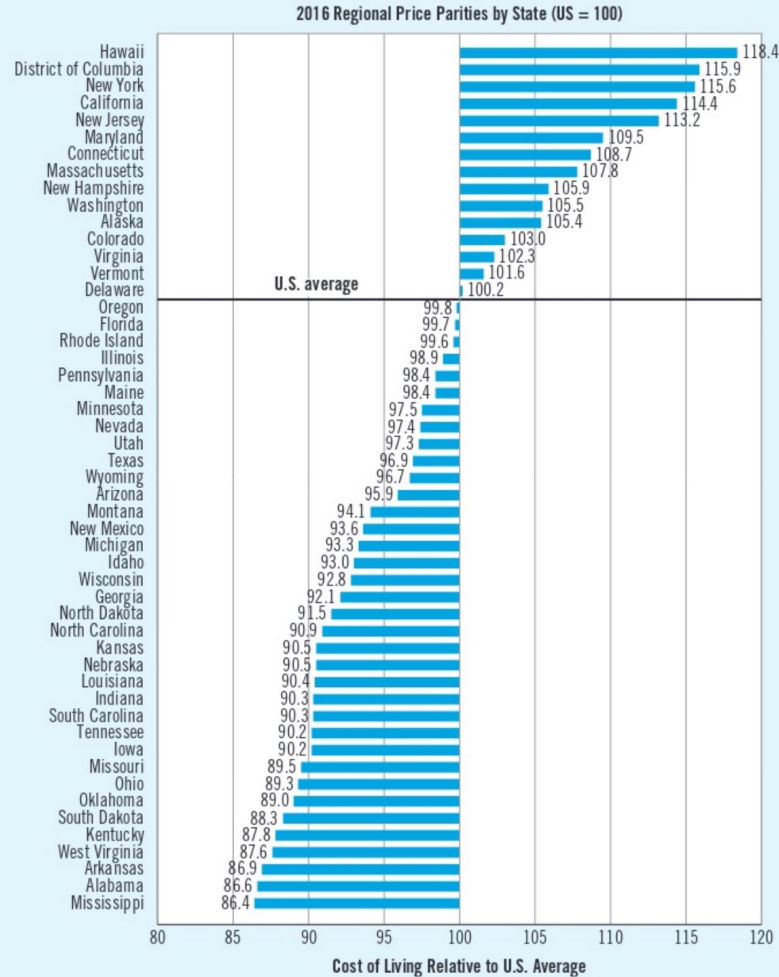
*The cost of living changes across time and across geography*

**FIGURE 3**

## Regional Variation in the Cost of Living

This figure shows how the costs of living in the 50 U.S. states and the District of Columbia compare to the U.S. average.

Source: U.S. Department of Commerce.



Variation in the cost of living across locations.

Why could this matter for a firm thinking of building a new manufacturing facility?

Source: Mankiw (2018, p. 219)



### IMPORTANT CONSIDERATION III

## Real versus nominal interest rates

### Driving question

**What is left of a quoted rate after you subtract inflation?**

## MAIN TAKEAWAYS

# What to carry out of this meeting

### Driving question

**A price index is a tool. Know what it prices — and what it misses.**

# Main takeaways — CPI measurement and limitations

## **Basket**

CPI tracks the cost of living via a fixed basket and the inflation rate that follows from it.

## **Steps**

Fix basket weights, gather prices, use a base-year index, derive inflation.

## **Issues**

Substitution bias, new goods, and quality change often overestimate inflation.

## **Alternatives**

GDP deflator, PCE, and Truflation.

## Main takeaways — applications and implications

- 01** Use CPI to adjust historical dollar values (real vs nominal costs — e.g. college tuition).
- 02** Indexation corrects contracts and taxes for inflation.
- 03** Regional price variations affect business decisions (e.g. Amazon compensation if it opens a headquarters outside Seattle).
- 04** Real interest rates (nominal minus inflation) guide borrowing and saving.
- 05** Business relevance: pricing, wages, and risk management when inflation varies.

# Important consideration III — real vs nominal interest rates

Book Content

Interest Rates  
(percent per year)



**FIGURE 4**

## Real and Nominal Interest Rates

This figure shows nominal and real interest rates using annual data since 1965. The nominal interest rate is the rate on a three-month Treasury bill. The real interest rate is the nominal interest rate minus the inflation rate as measured by the CPI. Notice that nominal and real interest rates often do not move together.

Source: U.S. Department of Labor; U.S. Department of Treasury.

The two series tend to move together, but...

Think about this as an investor.

What is the difference if you have liquidity (save) versus if you do not (borrow)?

What would be alternatives?